

CSHTML Intelligence

Complete Extraction & Reverse Engineering Reference

A Comprehensive 21-Dimension Analysis Framework

Multi-Agent Architecture • Semantic Intelligence • Code Generation

Generated: 2026-03-14

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Right-click and select 'Update Field' to refresh page numbers

Executive Summary

This document consolidates comprehensive intelligence extraction methodology for ASP.NET CSHTML/Razor views, enabling automated reverse engineering of legacy enterprise applications into modern technology stacks.

The framework extracts 350+ intelligence points across 21 dimensions, transforming static view files into actionable development artifacts including database schemas, API specifications, UI components, and test suites.

Part 1: The 21-Dimension Intelligence Framework

Each dimension captures a specific category of intelligence from CSHTML files, producing structured metadata suitable for code generation, documentation, and project planning.

Dimension 1: Page Metadata Intelligence

Extracts fundamental page-level information including view model bindings, layout references, controller associations, and permission requirements.

Metadata Field	Extraction Method
View Model	@model Directive → DTOPermission
Page Title	ViewBag.Title = 'Organization'
Layout Template	Layout = '~/Views/Shared/_Default.cshtml'
Controller	Html.BeginForm('Organization', 'Organization')
Form Method	FormMethod.Post, enctype='multipart/form-data'

Table 1: Page Metadata Extraction Examples

Dimension 2: Form Intelligence

Comprehensive form analysis including nested forms, submission targets, and multi-form coordination patterns.

Create Form Configuration

Property	Value	Intelligence Impact
Form ID	OrganizationForm	JavaScript validation binding
Action URL	/Organization/ InsertOrganization	API endpoint generation
Enctype	multipart/form-data	File upload support required
Default Button	SubmitOrganization	Form submission trigger mapping

Dimension 3: Field Intelligence

Each input field is analyzed for semantic type, validation requirements, and UI component mapping. The extraction yields 22 create form fields and 14 update form fields.

Create Form Fields Summary

#	Field ID	Name	Type	MaxLen	Required
1	Code	code	text	2	Yes
2	IsActive	isactive	checkbox	—	No

#	Field ID	Name	Type	MaxLen	Required
3	Name	name	text	100	Yes
4	OrganizationTypeId	OrganizationTypeId	select	—	No
5	orgEmail	Email	text	100	Yes
6	UAN	UAN	text	16	No
7	TelNo	telno	text	15	Yes
8	CellNoOne	cellnoone	text	15	Yes
9	CountryId	CountryId	select	—	Yes
10	ProvinceId	ProvinceId	select	—	Yes

Dimension 4: Database Schema Inference

From the CSHTML field analysis, database schema is inferred including primary keys, foreign keys, data types, and constraints.

Primary Table: Organization

Key Inference: ID type = UNIQUEIDENTIFIER based on JavaScript evidence: organizationId: '@Guid.Empty'

Column	Data Type	Constraints	Nullable	FK	Default
Id	UNIQUEIDENTIFIER	PK, DEFAULT NEWID()	No	—	NEWID()
Code	NVARCHAR(2)	NOT NULL, UNIQUE	No	—	—
Name	NVARCHAR(100)	NOT NULL, UNIQUE	No	—	—
OrganizationTypeId	INT	NULL	Yes	OrganizationTypes	—
Email	NVARCHAR(100)	NOT NULL, UNIQUE	No	—	—
CountryId	INT	NOT NULL	No	Countries	—
IsActive	BIT	NOT NULL	No	—	1

Part 2: Column Intelligence Engine

The Column Intelligence Engine provides 4-tier semantic type inference from SQL column definitions and naming patterns.

Tier 1: Data Type Based Inference (100% Reliable)

SQL Type	UI Component	Validation
BIT	Toggle / Checkbox	Boolean
INT / BIGINT	Number Input	Integer only
DECIMAL / MONEY	Currency Input	2 decimal places
DATETIME / DATE	Date Picker	Date format
UNIQUEIDENTIFIER	Hidden / Auto-generated	UUID format
NVARCHAR(MAX)	Textarea	Max length

Tier 2: Name Pattern Based Inference (90% Reliable)

Name Pattern	Inferred Type	UI Component	Notes
Email	Email	Email Input + validation	Shows @ keyboard
Password, Pin	Password	Password Input	Never display
Phone, Cell, Tel	Phone	Phone Input + mask	Country aware
IsActive, Is*	Boolean Flag	Toggle Switch	Filter-worthy
Name, Title	Display Name	Text Input	Primary field
*Id (FK)	Foreign Key	Dropdown / Search	From referenced table
Amount, Price	Currency	Currency Input	Right-aligned

Part 3: API Endpoint Intelligence

API endpoints are derived from form actions, AJAX calls, and stored procedure analysis.

Detected API Endpoints

#	Method	URL	Purpose
1	POST	/Organization/InsertOrganization/	Create organization + admin user
2	POST	/Organization/UpdateOrganization	Update organization
3	POST	/Organization/GetOrganizations	List all organizations
4	GET	/Organization/GetStateOrProvinces	Get provinces by country
5	GET	/Organization/GetCities	Get cities by province
6	POST	/Organization/CheckOrganizationAvailability/	Validate name uniqueness

Part 4: Business Rule Intelligence

Business rules are extracted from validation logic, stored procedures, and UI workflow patterns.

#	Business Rule	Evidence
1	Organization Code must be exactly 2 digits	minlength='2' maxlength='2'
2	Organization Code cannot be changed after creation	readonly on update form
3	Organization Name is unique	Remote validation + error codes
4	Admin user created with organization	Single AJAX creates both
5	Code uniqueness enforced server-side	Error code -1 returned
6	Soft delete pattern (IsActive)	IsDeleted=0 in SP queries

Part 5: Validation Intelligence

34 validation rules extracted from bootstrapValidator configuration, HTML attributes, and server-side error codes.

Validation Matrix Sample

Field	Validator	Rule/Pattern	Error Message
code	notEmpty	Required	Enter Organization Code
code	digits	Numbers only	Only digits allowed
code	remote	AJAX uniqueness	Code Already Exists
name	regexp	^[A-Za-z0-9_/#&+:-"()]+\$	Invalid characters
Email	regexp	Email pattern	Valid Email required
User.password	identical	Match confirmpassword	Passwords don't match

Part 6: Multi-Agent Architecture

The platform uses a hierarchical agent system where specialized agents collaborate through shared context and message-based communication.

Agent Hierarchy

Orchestrator (Master Agent) coordinates three layers:

- **Schema Layer:** SQL Parser, SP Parser, View Analyzer, CSHTML Parser
- **Intelligence Layer:** Column Intelligence, FK Resolver, Dependency Graph
- **Generation Layer:** Prisma, API, Screen, Test, Documentation generators

Agent Communication Protocol

Agents communicate through a shared context object, never directly. The orchestrator decides execution order based on dependencies.

Part 7: Stored Procedure Intelligence

Stored procedures reveal behavior and business logic that table DDL alone cannot capture. The SP parser extracts parameters, table access patterns, and workflow sequences.

SP Naming Pattern Classification

Pattern	Inferred Operation	Generates
SP_DDL_*	Dropdown/Lookup data source	React <Select> component
SP_Get* / SP_Fetch*	Read/Query operation	List/Detail page
SP_Add* / SP_Create*	Create operation	Create form + API POST
SP_Update* / SP_Save*	Update operation	Edit form + API PUT
SP_Delete*	Delete operation	Delete confirmation
SP_Report*	Reporting query	Report page + export

Part 8: Strategic Implementation Recommendations

Build Order (Revised)

Based on maximum user value delivery:

- 1. **Column Intelligence Engine** — THE differentiator, no other tool does this
- 2. **Screen Blueprint Generator** — time-saver, demo-worthy feature
- 3. **AI Question Engine** — engagement driver, makes tool feel intelligent
- 4. **Stored Procedure Parser** — highest value-add for migration scenarios

Offline-First Architecture

90% of 'AI' features are actually rule-based pattern matching and can work 100% offline. This is a competitive advantage for healthcare, government, and enterprise clients with data sensitivity concerns.

Feature Offline Compatibility

Feature	Needs LLM?	Reality
SQL Parser	No	Pure regex pattern matching
Column Intelligence	No	Rules-based inference engine
ERD Generator	No	Mermaid template generation
Prisma Generator	No	Template transformation
Screen Blueprint	Optional	95% rules, LLM optional
User Story Generator	Optional	Template expansion

Conclusion

This comprehensive intelligence extraction framework enables automated reverse engineering of legacy ASP.NET applications into modern technology stacks. The 21-dimension analysis produces 350+ intelligence points, transforming static view files into actionable development artifacts.

Key differentiators of this approach:

- **Multi-source analysis:** SQL + SPs + Views + CSHTML → unified intelligence
- **Semantic understanding:** Not just structure, but meaning
- **Offline-first:** 85% of features work without AI
- **Enterprise-ready:** Healthcare compliance, multi-tenant architecture

The system transforms legacy codebases from liabilities into documented, understood, and migratable assets.